

Kevin Shu

kevinshu.me

Work

July 2026-Present **Assistant Professor** (Combinatorics and Optimization),
University of Waterloo, Waterloo, ON

2024-July 2026 **Postdoctoral Researcher** (Computing and Mathematical Sciences),
California Institute of Technology, Pasadena, CA

Education

2019-2024 **PhD.**, Algorithms Combinatorics and Optimization,
Georgia Institute of Technology, Atlanta, GA

- Advised by Grigoriy Blekherman in the Department of Mathematics.

2014-2018 **B.S.**, Mathematics, Computer Science,
California Institute of Technology, Pasadena, CA

As is common in mathematics, authors are typically listed alphabetically, and the ordering does not reflect the contributions of the coauthors.

Publications

Beyond Minimax Optimality: A Subgame Perfect Gradient Method (with Ben Grimmer and Alex Wang), Mathematical Programming, 2026.

Debiasing Polynomial and Fourier Regression (with Chris Camaño and Raphael A. Meyer), SOSA, 2026.

Composing Optimized Stepsize Schedules for Gradient Descent (with Ben Grimmer and Alex L. Wang), Mathematics of Operations Research, 2025.

A Semidefinite Hierarchy for the Expected Independence Number of a Random Graph (with Diego Cifuentes, and Alejandro Toriello), Optimization Letters, 2025

Accelerated Objective Gap and Gradient Norm Convergence for Gradient Descent via Long Steps (with Ben Grimmer and Alex L. Wang), Informs Journal on Optimization, 2025

Symmetric Hyperbolic Polynomials (with Greg Blekherman and Julia Lindberg), Journal of Pure and Applied Algebra, 2025

Hidden convexity, optimization, and algorithms on rotation matrices (with Akshay Ramachandran and Alex L. Wang), Mathematics of Operations Research, 2024

Linear Principal Minor Polynomials: Hyperbolic Determinantal Inequalities and Spectral Containment (with Greg Blekherman, Mario Kummer, Raman Sanyal and Shengding

Sun), International Mathematics Research Notices, 2022

Hyperbolic Relaxation of k -Locally Positive Semidefinite Matrices (with Grigoriy Blekherman, Santanu Dey and Shengding Sun), SIAM Journal on Optimization, 2022

Sums of Squares and Sparse Semidefinite Programming (with Grigoriy Blekherman), SIAM Journal for Applied Algebra and Geometry, 2021

Syntactic Structures and Code Parameters (with Matilde Marcolli), Mathematics in Computer Science, 2017

Preprints

A Unified Theory of H -Duality in First-Order Methods (with Alex Wang), arXiv:2609.03281, 2026

Accelerated Gradient Descent via Long Steps (with Ben Grimmer and Alex Wang), arXiv:2309.09961, 2023

Quadratic Programming with Sparsity Constraints via Polynomial Roots, arxiv:2208.11143, 2022

Talks and Presentations Given

Hidden Convexity, presented at ICCOPT 2025

Hidden Convexity and the Rotation Group, presented at the SIAM Conference on Applied Algebraic Geometry 2025

Algebraic Methods in Convex Optimization, presented at the UCLA Math Colloquium 2025

Semialgebraic Methods in Convex Optimization, presented at the Joint Mathematics Meetings 2025

Hidden convexity, optimization, and algorithms on rotation matrices, presented at the Informs Optimization Society conference 2024

Symmetric Hyperbolic Polynomials, presented at the SIAM Conference on Applied Algebraic Geometry 2023

Sparse Regression and PCA via Polynomial Roots, presented at the SIAM Conference on Optimization 2023

Hyperbolicity Cones and Sparse Optimization, presented at the MIT LIDS seminar

2023

Symmetrically Hyperbolic Polynomials, presented at the Oberwolfach Meeting on New Directions in Algebraic Geometry

2023

Sparse Quadratic Programs via Polynomial Roots, presented at the Carnegie Mellon University ACO seminar

2023

Sparse Quadratic Programs via Polynomial Roots, presented at the Centrum Wiskunde and Informatica Networks and Optimization seminar

2022

Approximating Sparse Semidefinite Programs, presented at the INFORMS conference

2021

Poster on Sparse Semidefinite Programs, presented at the MIP and IPCO conferences

2021

Causal Inference and Optimization, presented at the ACO Student Seminar

2021

Lightning Talk on Hyperbolic Relaxations of Locally-PSD Matrices, presented at the ICERM - Symmetry, Randomness, and Computations in Real Algebraic Geometry.

2020

Academic Honors

2025 Best Thesis Award for the Georgia Tech Mathematics Department

2022 ACO-ARC Fellowship

2022 ARCS Foundation award

2021 Honorable Mention at the MIP Conference Poster Competition

2021 Honorable Mention at the IPCO Conference Poster Competition

2021 David L. Brown Fellowship from the Georgia Tech Math Department

2018 National Science Foundation Graduate Research Fellowship Recipient

2018 Georgia Institute of Technology President's Fellowship Recipient

Conference Organization

Coorganizer for the Session on Algebraic Methods in Optimization, ICCOPT July 2025

Coorganizer for the Georgia Tech Student Algebra Seminar, Georgia Tech, GA

August 2022-December 2023

Coorganizer for the Special Session on Convexity, SIAM Conference on Applied Algebraic Geometry, Georgia Tech, GA

July 2023

Organizer for the AMS Special Session on Algebraic Methods in Algorithms, Spring 2023 Southeastern Section Meeting of the AMS, Georgia Tech, GA

March 2023

**Research
Experience**

Visiting Scholar, Max-Planck Institute for Mathematics in the Sciences, Leipzig, Germany
Summer 2022

- Working under the supervision of Rainer Sinn and Bernd Sturmfels.

Research Assistantship, Georgia Tech, Atlanta, GA
Summer 2020

- Funded in part by NSF grant DMS-1901950 and the ACO department.
- Advised by Grigoriy Blekherman.

**Outreach and
Community
Service**

Representative for the Diversity, Equity, and Inclusion committee, Georgia Tech, GA
2022-2023

First Year Mentor, Georgia Tech, Atlanta, Georgia
2020-2021

Directed Reading Program Mentor, Georgia Tech, Atlanta, Georgia
2020-2021

Senior Class President, Caltech, Pasadena, CA
2018-2019

Board of Control Secretary, Caltech
2017

**Teaching
Experience**

Differential Equations Teaching Assistant, Georgia Tech, Atlanta, GA
Aug 2022-Dec 2022

Differential Equations Teaching Assistant, Georgia Tech, Atlanta, GA
Aug 2021-Dec 2021

Number Theory Lecture Assistant, Georgia Tech, Atlanta, GA
Jan 2021-May 2021

Differential Equations Teaching Assistant, Georgia Tech, Atlanta, GA
Jan 2020-May 2020

Linear Algebra Teaching Assistant, Georgia Tech, Atlanta, GA
Aug 2019-Dec 2019

Advanced Algorithms Teaching Assistant, Caltech, Pasadena, CA
Jan 2018-Mar 2018

Linear Algebra Teaching Assistant, Caltech, Pasadena, CA
Sep 2017-Dec 2017

Introduction to Algorithms Teaching Assistant, Caltech, Pasadena, CA
Jan 2017-Mar 2017

**Work
Experience**

Full-time Software Engineer, Google, Mountain View, CA
August 2018-July 2019

- Full stack web development for a data labelling service (Crowd-Compute)

- Lead an initiative to update authentication/authorization to more modern technologies.
- Added a major feature for tracking work in the system.
- Managed production releases and infrastructure issues.
- Worked with C++, Java.

Software Engineering Intern, Google, Mountain View, CA

Aug 2018-Jul 2019

- Gathered data from online sources by parsing Reddit pages.
- Built a machine learning model to provide movie recommendations.
- Worked with C++, Python.